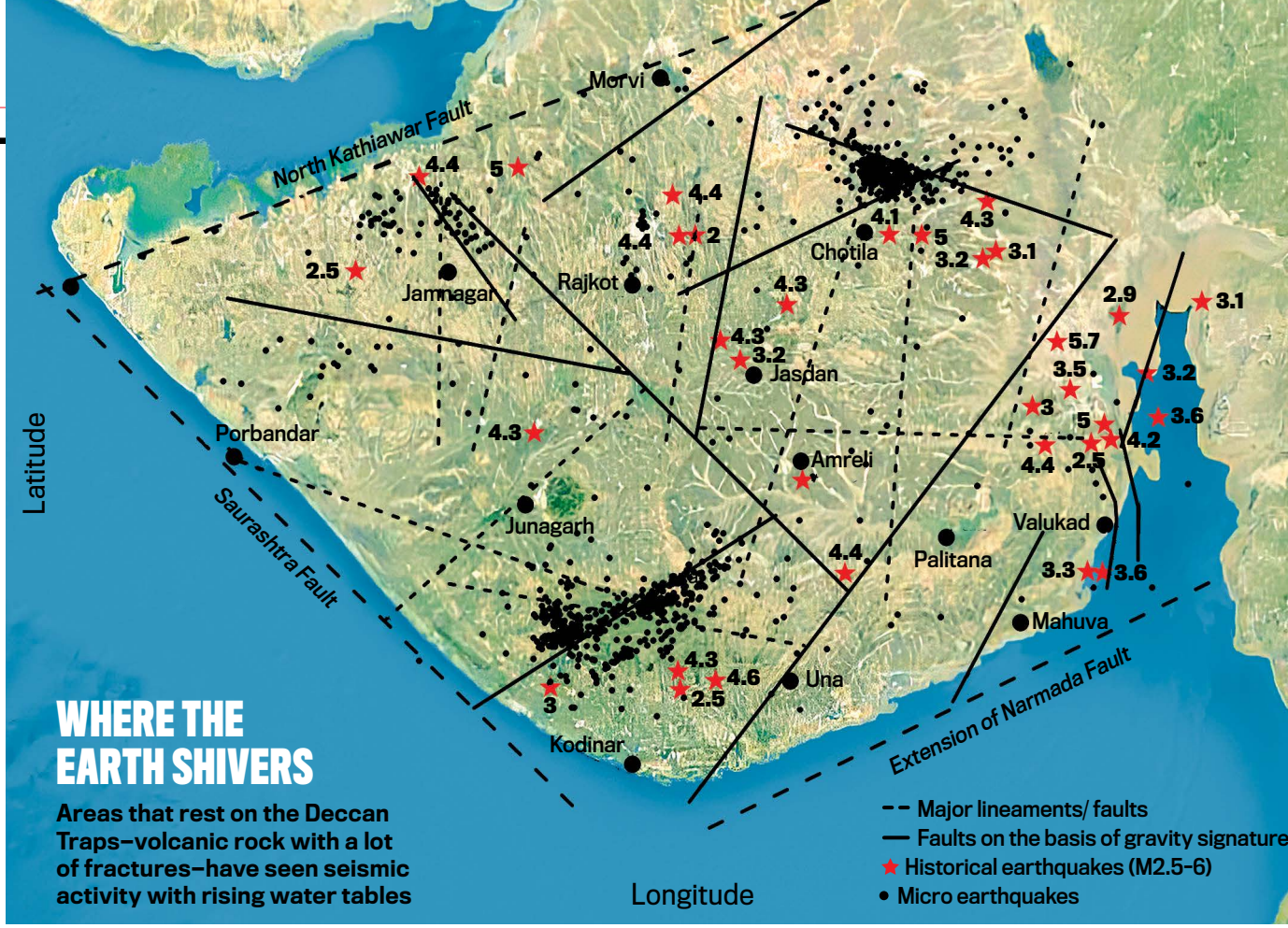


Source: Institute of Seismological Research, Gandhinagar; map not to scale

causing these swarms hardly have lengths of a few kilometres. "Gases and highly charged particles are released, which keeps the pressure on. But Random studies of the region have revealed, to our relief, that radon gas, which could signal a bigger quake, is not released here," adds Chopra.

Wherever swarm activity is witnessed, the region has lots of fractures that run deep. Except for Kutch, most of Gujarat lies in what is classified as Zone 3, as per seismic zoning map prepared by the Bureau of Indian Standards, which indicates moderate seismic activity and no major fault line. Kutch, meanwhile, is in the highest seismic risk Zone 5, which led to the massive earthquake in 2001 and the Allah Bund (Dam of God) earthquake of 1819 that entirely changed the topography of the region. Kutch is about 300 km from the collision points of the Indian and Arabian tectonic plates, which continue to be an active plate boundary.

If this seismic activity does not necessarily portend a major quake, why is the ISR investing so much resources and monitoring this activity so closely? "Understanding seismic conditions is the foundation of any development plan. Be it the construction of ports, dams, cities, tall buildings or industrial activity, everything depends on the seismic behaviour of that region. It determines the kind of material to be used for construction, the height and the load bearing of buildings," Chopra says. Constituted after the 2001 Bhuj earthquake, ISR has a dense coverage of 200 monitoring stations in Gujarat.

What makes these earthquakes sinister for villagers in Saurashtra is not that the earth continues to shake intermittently, but it does so with an audible rumble. These quakes are associated with an audible frequency range of 20 Hz-20 kHz. Residents of Khankotda and Matwa villages in Jamnagar reported with great fright that the earthquakes generate a blasting kind of sound. "The rumble is a sign that it is a shallow quake," Chopra explains.

SEVERAL REGIONS HAVE BEEN EXPERIENCING SMALL QUAKES THAT KEEP RECURRING

Sarpanch Mansukh Molariya of Mitiyala village in Amreli, which has been the epicentre of these swarm earthquakes, says, "The rumbling disturbed the villagers immensely. Over time, some of us have reconciled to these earthquakes after scientists, the district collector and the MLA assured us this was not the precursor to a big quake. But several families migrated out of the village in the initial months." Even now, some families, out of fear, prefer to sleep in the open. However, that too is fraught with risk as the village borders the forest and the Asiatic Lions roam the region freely. "But our village women and children fear earthquakes more than a lion attack," Molariya says.

Chopra sums it up matter-of-factly: "You cannot avoid earthquakes. We have to understand how to deal with them." And that is easier said than done. ■